

Design AX Brief

A self-running daily news brief for a design organization's AI transformation. Six AI subagents collect, verify, curate, write, illustrate, and publish 5 fresh design×AI news cards into a single static web page — every day.

1. What it is

Design AX Brief is a static, dependency-light web page (`Design AX Brief.html`) that shows today's 5 most useful design×AI news cards in a hero carousel, plus a "weekly deck" of the previous days. It is generated by a **6-agent pipeline** orchestrated by a single skill, `/design-ax-daily-news` . The source of truth is one JSON file (`pipeline/news_data.json`); everything else is derived from it.

- **Fresh & verified.** Only news whose *real published time* is within the last 72 hours is shown.
- **No duplicates.** Each distinct story appears on exactly one date (earliest-wins), judged by URL + content.
- **No padding.** If fewer than 5 fresh, non-duplicate stories exist, it publishes what genuinely exists.
- **Calm, minimal Korean editorial voice;** each card carries a story-specific in-article image.

2. How it works — the 6-agent pipeline

Running `/design-ax-daily-news` dispatches six subagents **strictly in order**. Each writes one artifact under `pipeline/` ; the next reads it. After each stage the orchestrator verifies the artifact exists and is non-empty before continuing.

```
keyword_pool.json → 1 ax-planner → keywords.json
keywords.json → 2 ax-librarian → candidates.json (~15-20, time-verified)
candidates.json → 3 ax-curator → selected.json (top 5, deduped)
selected.json → 4 ax-writer → cards.json (Korean copy)
cards.json → 5 ax-media → media.json + images
cards + media → 6 ax-publisher → axbrief-data.js + archive
```

AGENT	ROLE	KEY RULES
ax-planner	Plan today's search keywords.	Always uses <code>core</code> keywords; rotates a date-based subset of the broad <code>keyword_pool.json</code> ; ~20–28 queries; avoids topics already shown.
ax-librarian	Search the web & collect candidates.	For each candidate, fetches the article and verifies the <i>real</i> published time (og / JSON-LD / byline). Gates with <code>freshness.py</code> (flat 72h). Drops anything undateable.
ax-curator	Select the best 5.	Scores actionability / novelty / credibility / diversity. Drops duplicates vs the last 5 days (URL + content, not image) and backfills the next-ranked candidate.
ax-writer	Write calm Korean card copy.	1-line headline (balanced 2-line break), body that fills ≤3 lines (≤110 chars). No invented facts.
ax-media	Give each card a distinct visual.	Prefers the in-article lead image (between title & body) → og:image → generated SVG. md5-compares so no two cards share a thumbnail.
ax-publisher	Publish into the live page.	Rolls yesterday into the deck (<code>roll.py</code>), regenerates <code>axbrief-data.js</code> (<code>build_data.py</code>), validates with <code>node --check</code> , archives the run.

3. The rules that keep it honest

Freshness — flat 72h

`pipeline/freshness.py` gates every candidate deterministically: an article is kept only if its verified publish time is within 72 hours of `now` (same window every weekday).

No duplicates (URL + content)

Two items are the same story if they share a URL/redirect, cover the same event via a different outlet, or are same-source same-topic posts. **A shared thumbnail is NOT a duplicate** — different articles are kept, and ax-media gives each a distinct in-article image.

Earliest-wins + backfill

A story collected on day N never reappears on N+1...N+5. If dropping a duplicate leaves a gap, the curator promotes the next-best candidate from the 15–20 pool.

Deterministic backstops

`roll.py` drops any rolled-in card whose URL already appears earlier; `build_data.py` **fails the build** if any two cards share a URL (and warns on a shared image).

Keyword pool & rotation. `pipeline/keyword_pool.json` holds a broad, categorized keyword set (AI/design tools, models, labs, people, generative media, robotics). The planner always searches the small `core` set and rotates a fresh subset of the larger pool each day — this widens coverage while keeping each day's search bounded.

4. The page

- **Hero carousel** — today's up-to-5 cards, swipeable, each with image, headline, body, and a source link.
- **Weekly deck** ("어제까지의 모든 소식") — the previous days as fanned mini-card stacks; hover a date to expand, click a card to open it large in the hero.
- **"오늘 소식으로" control** — appears while viewing a past day, returns the hero to today. (Today lives only in the hero, never duplicated into the deck.)
- Built with React + Babel from CDN; no build step, no server framework. Data & app are cache-busted per load so a fresh publish always shows.

5. Project structure

```
Design AX Brief.html      # the page (open this)
axbrief-app.jsx           # React UI (hero, deck, motifs) – Babel-compiled in browser
axbrief-data.js           # AUTO-GENERATED data (window.AX_NEWS + window.AX_DAYS)
.claude/
  skills/design-ax-daily-news/SKILL.md  # the /design-ax-daily-news orchestrator
  agents/ax-{planner,librarian,curator,writer,media,publisher}.md
pipeline/
  keyword_pool.json       # default search keywords (edit to customize)
  sources.json            # categories: label, accent color, motif, allowed domains
  news_data.json          # SOURCE OF TRUTH (today + last 5 days)
  freshness.py            # 72h recency gate (+ _test.py)
  roll.py                 # today -> deck, dedup (+ _test.py)
  build_data.py           # news_data.json -> axbrief-data.js (+ _test.py)
  media/                  # downloaded card images (tracked); deck/ = weekly banners
  runs/<date>/            # per-day archive of every artifact (git-ignored)
```

6. Clone & use it yourself

Prerequisites

- **Claude Code** (the CLI) — the skill and agents live under `.claude/` and run as subagents.
- **python3** and **node** (node only for the `--check` validation).
- A Chromium/Chrome browser for local preview & render verification.

Run the daily brief

```
git clone <your-fork> design-ax-brief
cd design-ax-brief
claude                # open Claude Code in the repo
> /design-ax-daily-news  # runs the full 6-agent pipeline for today
```

The skill asks for any extra search keywords (press enter to use the defaults), then runs all six stages and regenerates `axbrief-data.js`.

View the page

```
# serve over HTTP (file:// is blocked by CORS for the Babel/XHR app)
python3 -m http.server 8765
# open: http://localhost:8765/Design%20AX%20Brief.html
```

Customize

- **Keywords:** edit `pipeline/keyword_pool.json` — add tools / labs / people / concepts to `core` (always searched) or the categorized `pool` (rotated).
- **Categories & look:** edit `pipeline/sources.json` — each category's label, accent color, motif, and allowed source domains.
- **Voice / length:** tune `.claude/agents/ax-writer.md` (headline + body rules).
- **Window / cadence:** change the hours in `pipeline/freshness.py` (default 72h).

Tip — automate it. Point a daily scheduler (cron, or Claude Code's `/schedule`) at `/design-ax-daily-news` to publish a new brief every morning. Each run archives its full input/output under `pipeline/runs/<date>/` for reproducibility.

Deploy (optional)

It's a static site: commit `Design AX Brief.html`, `axbrief-app.jsx`, `axbrief-data.js`, and `pipeline/media/**` (image files are tracked), then host on GitHub Pages / Netlify / any static host. The intermediate pipeline JSON and `pipeline/runs/` are git-ignored.

Design AX Brief — generated guide. The page, pipeline, and agents are self-contained in the repo; this document describes the system as built. Source of truth: `pipeline/news_data.json`.